



西安交通大学
——金禾经济研究中心



Econometrics (I-1)

Weihong Zeng



Syllabus

- **Time:** Year 2026-2027, 1st semester
- 1-8th Week: Tuesday (3-4)&(7-8), 创新港5-5043
- **Lecturer:** 曾卫红
E-mail: zengwh@mail.xjtu.edu.cn
Office hour: Anytime by appointment (e-mail)
- **TA:** 刘龙翼 (博士生)
- **Platform:** 思源学习空间



Textbooks

- **Course Textbook (English):**
- Jeffrey M. Wooldridge, Introductory Econometrics---A Modern Approach **(Six Edition)**, South-Western, Thomson learning. 2016.
- A. Colin Cameron, Pravin K. Trivedi, Microeconometrics---methods and applications, Cambridge.2005.



Grading:

For my part:

	Contents	Points
1)	Attendance	5
2)	Homework	20
3)	Final Exam	60
4)	Final Report	15
Total		100

For whole course:

Teacher	Teaching hours	Weight
Zeng	32	1/2
Huang	8	1/8
Lin	24	3/8
Total		1

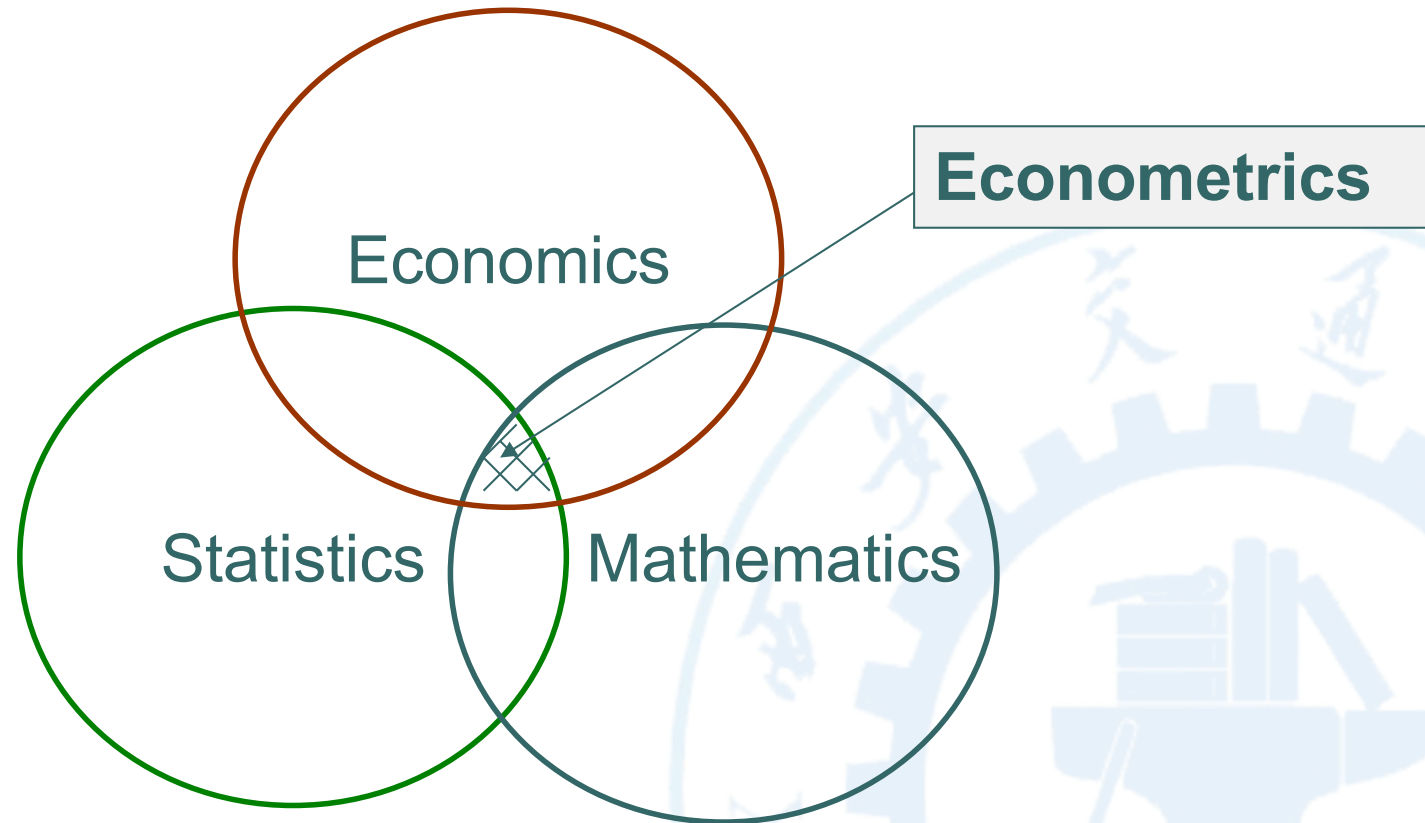
Note: We have homework every class, it is needed to submit up to the next Tuesday lecture.



•Attitude, not aptitude, decides your altitude.



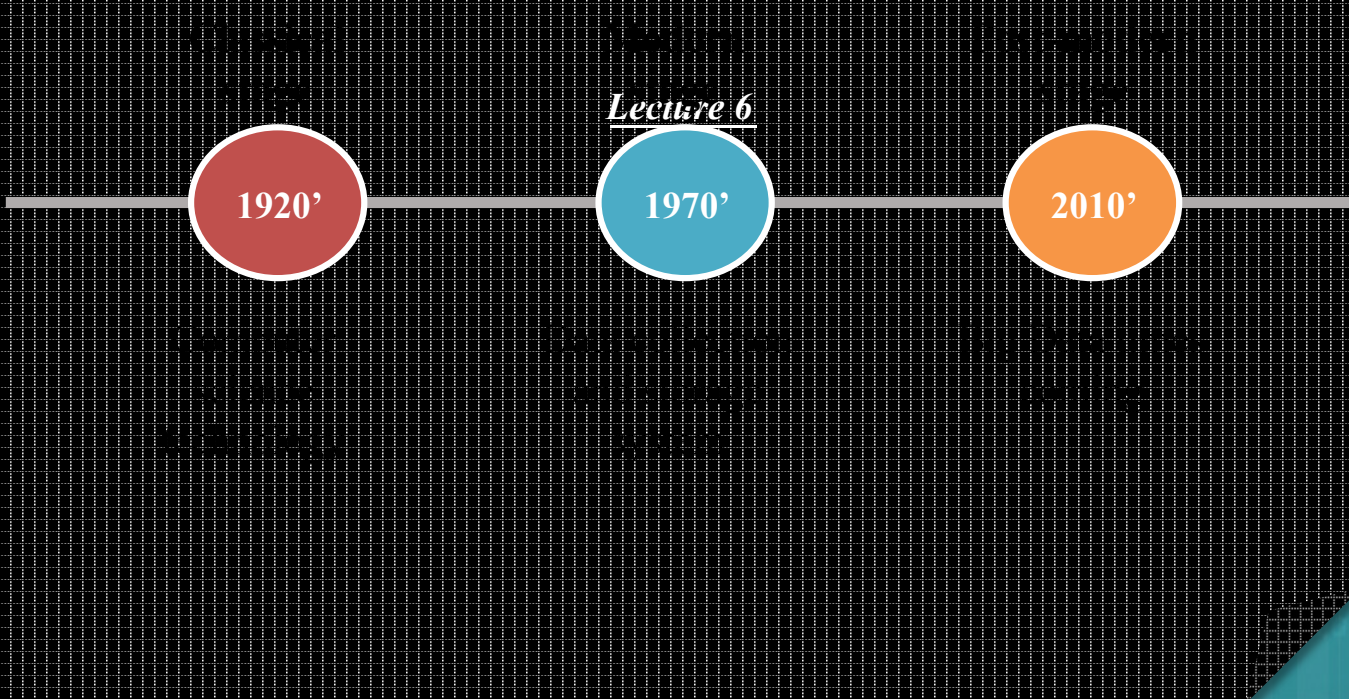
What is Econometrics?





- What is Econometrics?
- Econometrics consists of the ***application of mathematical statistics to economic data*** to lend empirical support to the models constructed by mathematical economics and to ***obtain numerical results.***
- Econometrics may be defined as ***the quantitative analysis of actual economic phenomena*** based on the concurrent development of theory and observation, related by ***appropriate methods of inference.***

Business Model Innovation





Development of Econometrics

○ A branch of Economics

- 20世纪20年代末30年代初诞生
 - 1926年挪威经济学家R.Frish提出Econometrics
 - 1930年成立世界计量经济学会
 - 1933年创刊《Econometrica》
- 20世纪40、50年代的大发展和60年代的扩张
- 20世纪70年代的批评和反思
- 20世纪70年代末以来非经典（现代）计量经济学的发展

计算机
科学
的
大
发
展

1896



△ The development of Econometrics ----From the owners of the Nobel Prize

- About 20% of the owners of the Nobel Prize in Economics is related with Econometrics filed:
 - 1969 R. Frish J. Tinbergen
 - 1973 W. Leotief
 - 1980 L. R. Klein
 - 1984 R. Stone
 - 1989 T. Haavelmo
 - 2000 J. J. Heckman D. L. McFadden
 - 2003 R. F. Engle C. W. J. Granger
 - 2011 Thomas J. Sargent, Christopher Sims
- About half of them experienced The President of the econometric society
- Over half of their achievements applied the theory of econometrics.



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The Bank of Sweden Prize in Economic Sciences in Memory of Alfred Nobel 1969

"for having developed and applied dynamic models for the analysis of economic processes"



Ragnar Frisch
Norway



Jan Tinbergen
the etherlands

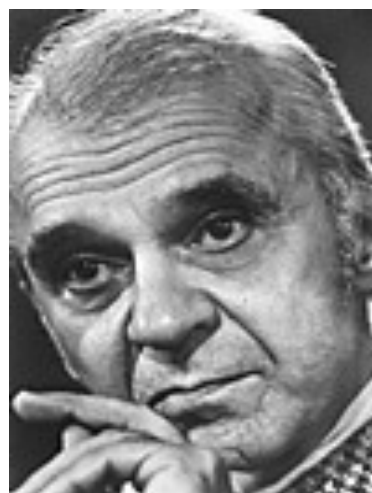


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The Bank of Sweden Prize in Economic Sciences in Memory of Alfred Nobel 1973

"for the development of the input-output method and for its application to important economic problems"



Wassily Leontief
USA



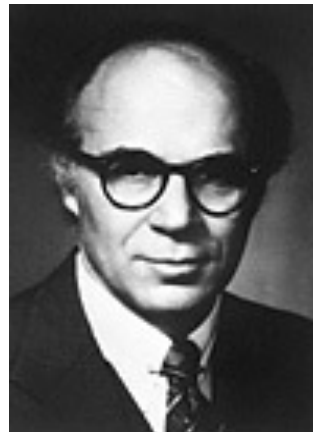


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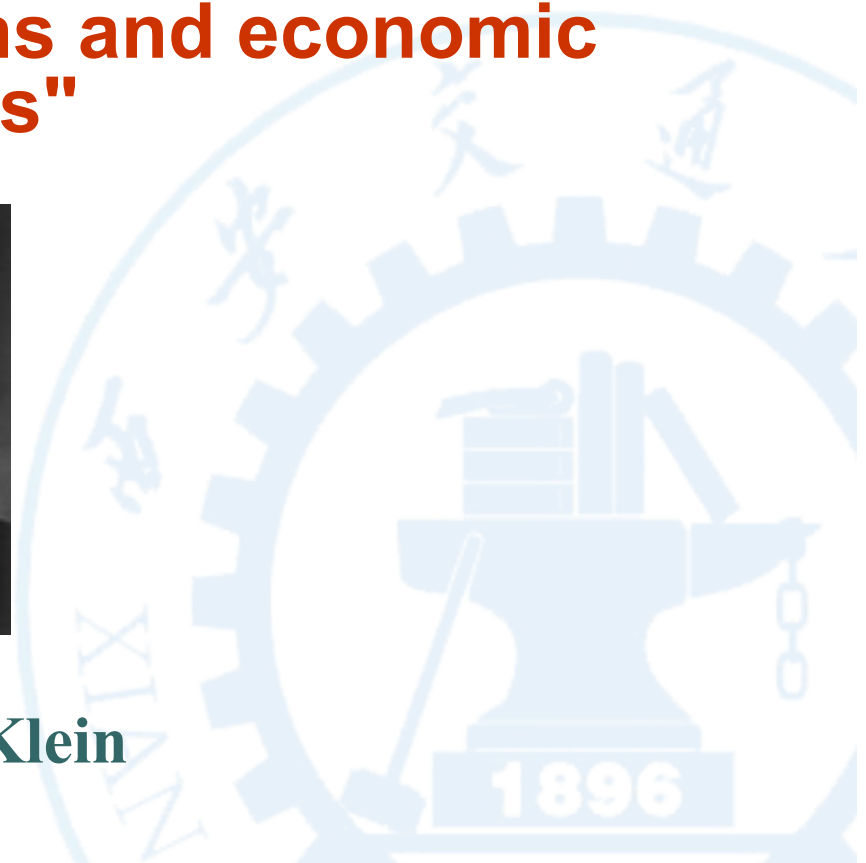


The Bank of Sweden Prize in Economic Sciences in Memory of Alfred Nobel 1980

"for the creation of econometric models and the application to the analysis of economic fluctuations and economic policies"



Lawrence R. Klein
USA





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The Bank of Sweden Prize in Economic Sciences in Memory of Alfred Nobel 1984

"for having made fundamental contributions to the development of systems of national accounts and hence greatly improved the basis for empirical economic analysis"



Richard Stone
Great Britain



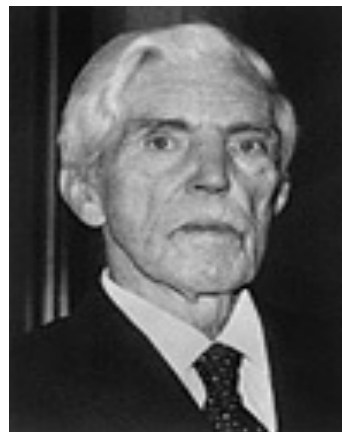


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The Bank of Sweden Prize in Economic Sciences in Memory of Alfred Nobel 1989

"for his clarification of the probability theory foundations of econometrics and his analyses of simultaneous economic structures"



Trygve Haavelmo
Norway





经典计量经济学

创立

Frisch

建立第1个应用模型

Tinbergen

建立概率论基础

Haavelmo

发展数据基础

Stone

发展应用模型

Klein

建立投入产出模型

Leontief

1896



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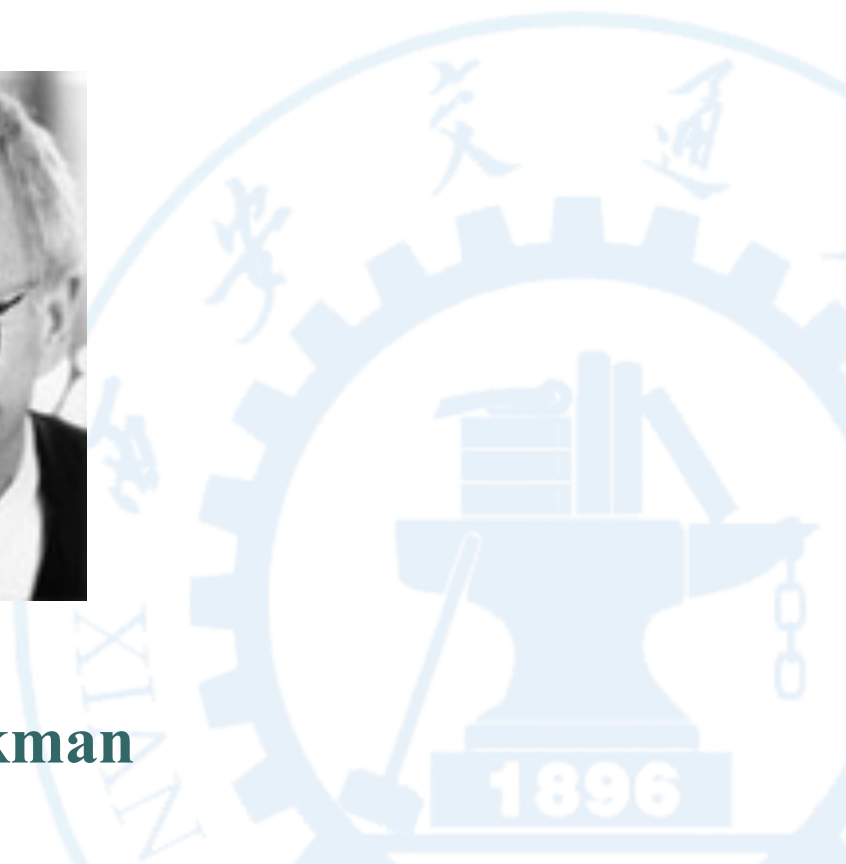


The Bank of Sweden Prize in Economic
Sciences in Memory of Alfred Nobel 2000

**"for his development of theory and methods
for analyzing selective samples"**



James J Heckman
USA





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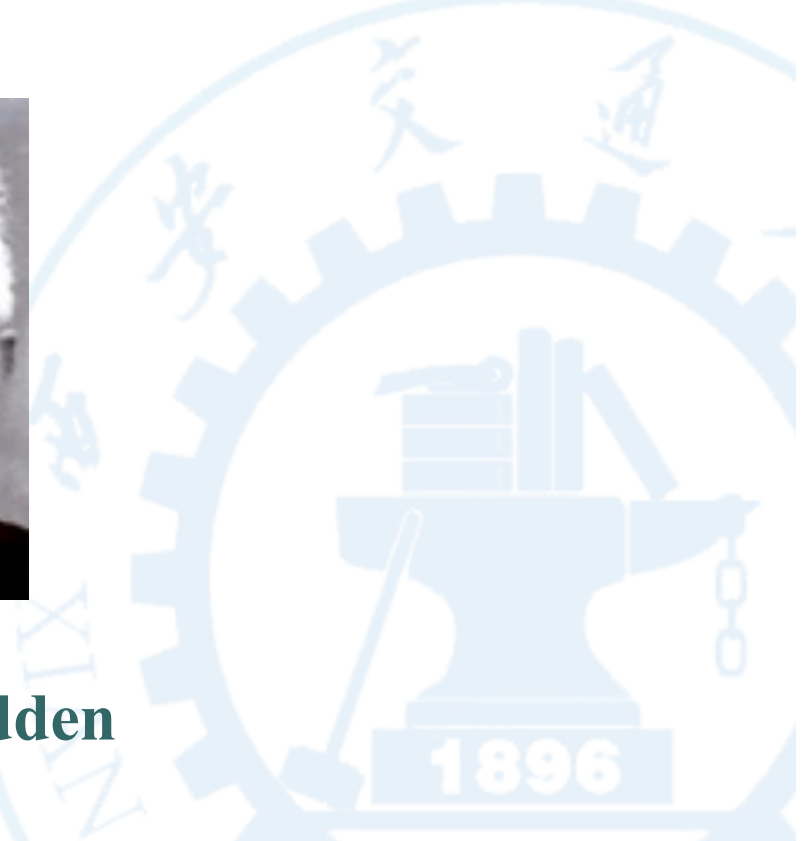


The Bank of Sweden Prize in Economic Sciences inMemory of Alfred Nobel 2000

"for his development of theory and methods for analyzing discrete choice"



Daniel L McFadden
USA





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The Bank of Sweden Prize in Economic
Sciences inMemory of Alfred Nobel 2003
**"for methods of analyzing economic time
series with common trends (cointegration)"**



Clive W. J. Granger
UK



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The Bank of Sweden Prize in Economic
Sciences inMemory of Alfred Nobel 2003
**"for methods of analyzing economic time
series with time-varying volatility (ARCH)"**



Robert F. Engle
USA



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The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2011

"for their empirical research on cause and effect in the macroeconomy"



Photo: U. Montan
Thomas J. Sargent
Prize share: 1/2



Photo: U. Montan
Christopher A. Sims
Prize share: 1/2

USA





非经典计量经济学

微观计量：
选择性样本模型

Heckman

微观计量：
离散选择模型

McFadden

时间序列：
协整理论—现代宏观计量

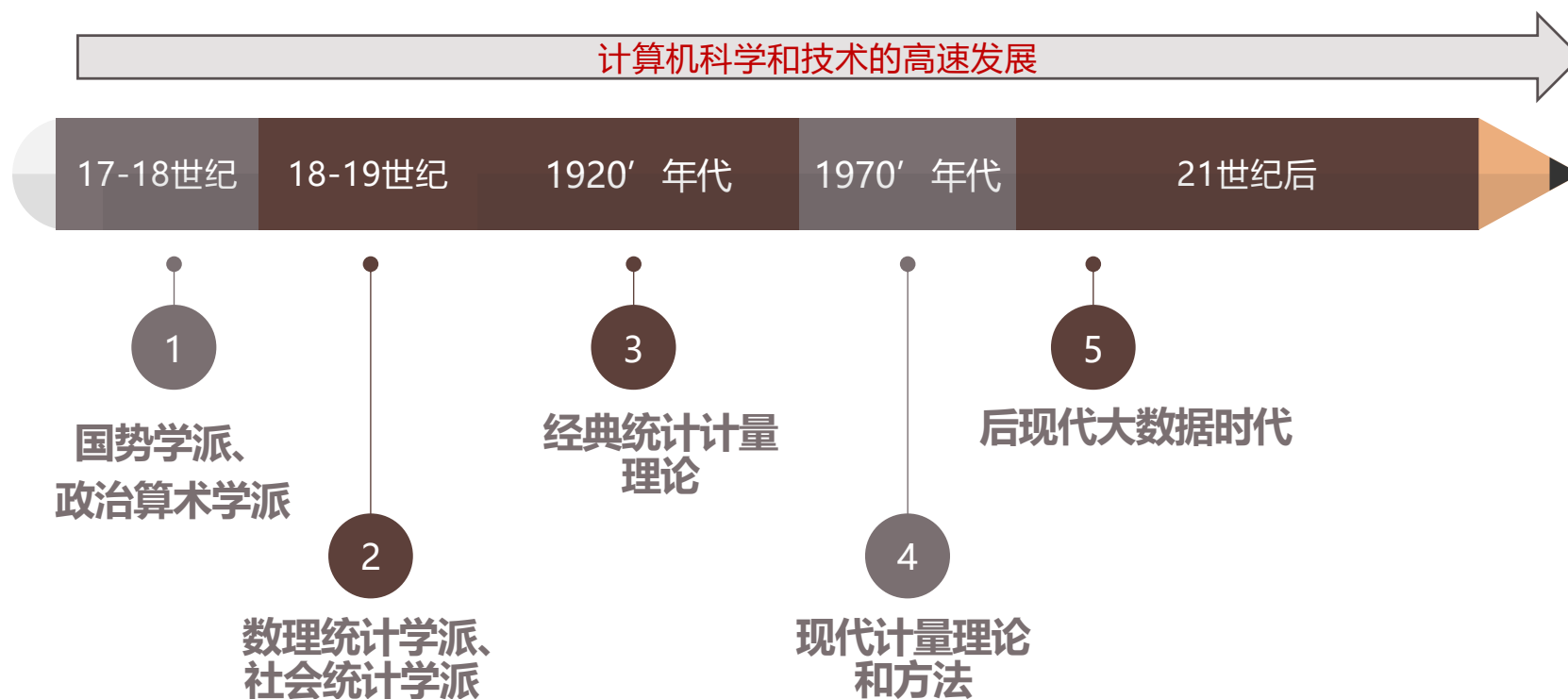
Granger

时间序列：
ARCH—现代金融计量

Engle

1896

应用统计学和计量经济学的前生后世



For more details for Nobel Prize in Economics, see:
<https://www.nobelprize.org/prizes/lists/all-prizes-in-economic-sciences/>

如何理解大数据、人工智能时代

1

大 大而不全 大而不深 大而不精

数据

风险激增 真假难辨 依赖算法

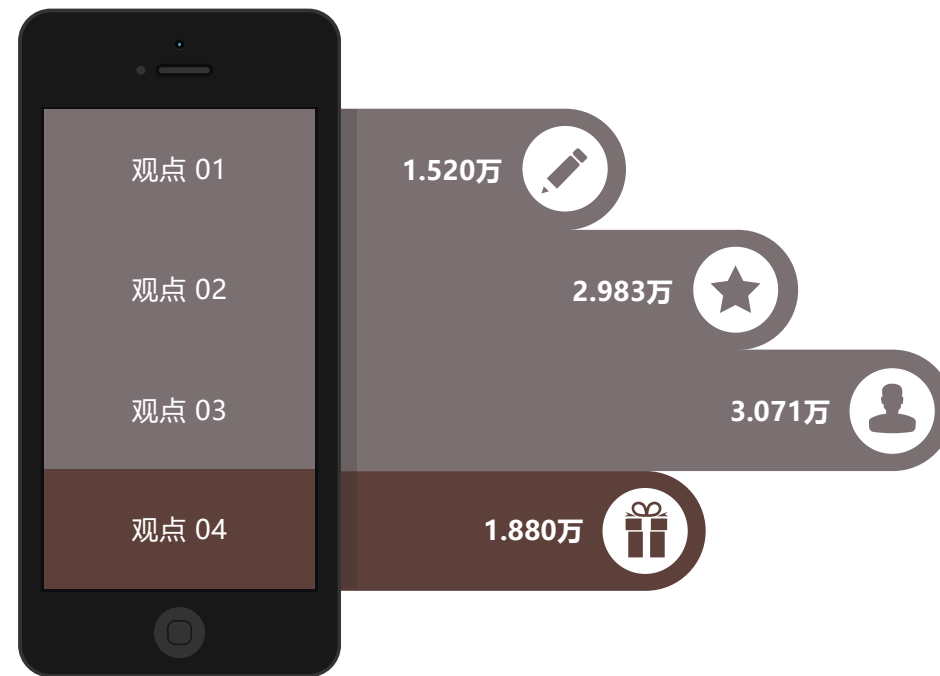
2

万物皆（数字）数据：文字、图像、音频、
视频、表情、心情.....

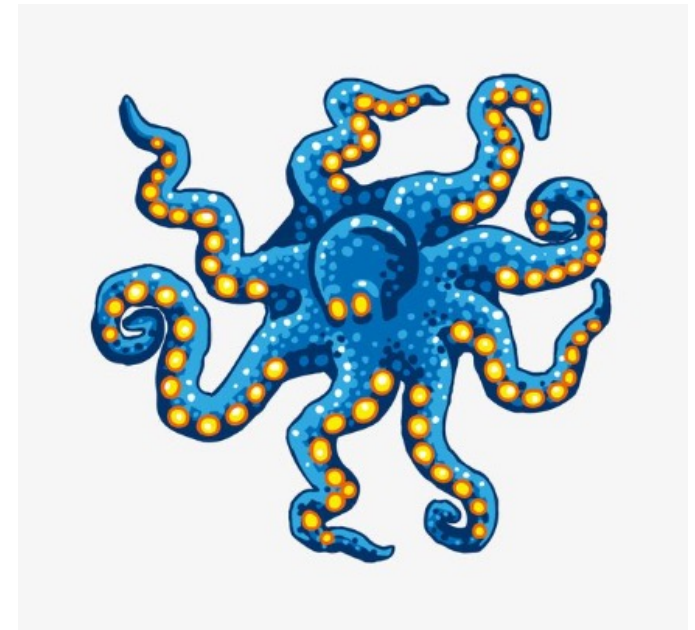
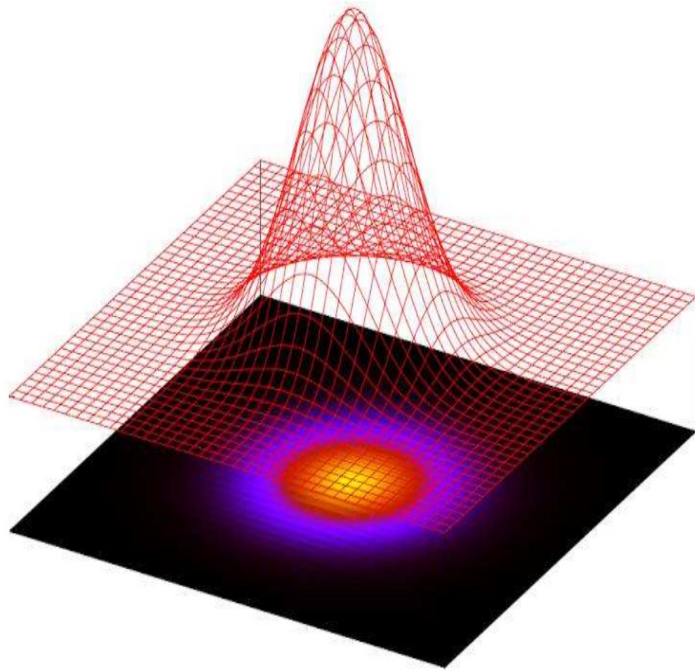
大数据时代：信息回声筒 Echo Chamber



Matthew Gentzkow, Clark Medalist 2014



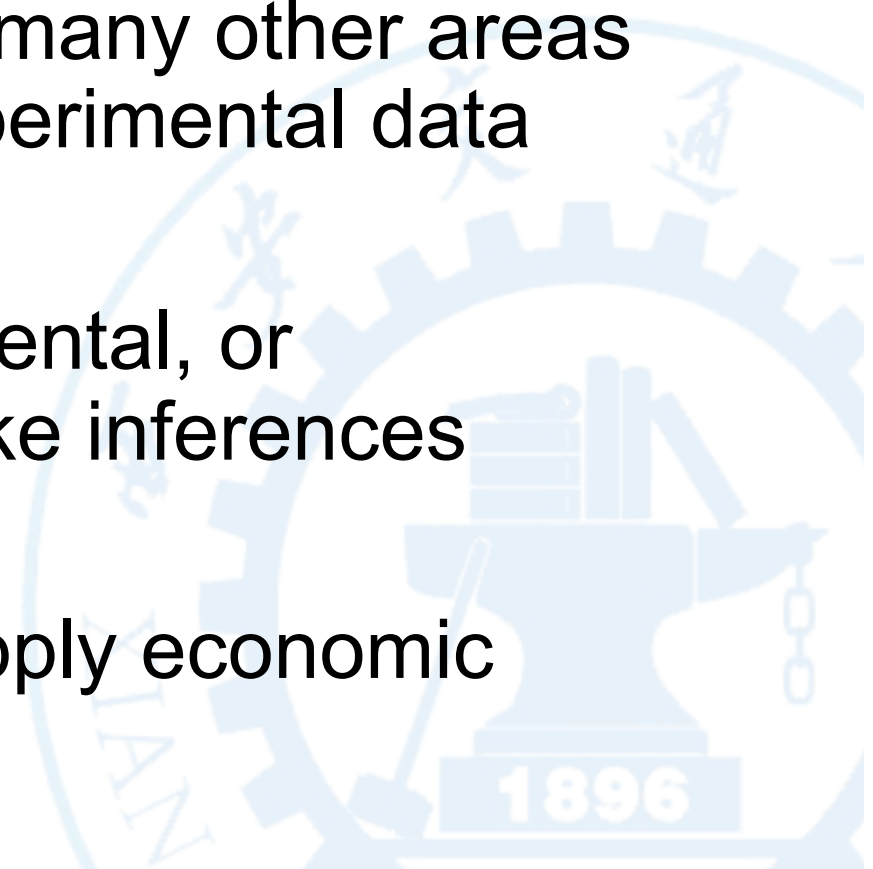
信息的集中与分散:





Why do we study econometrics?

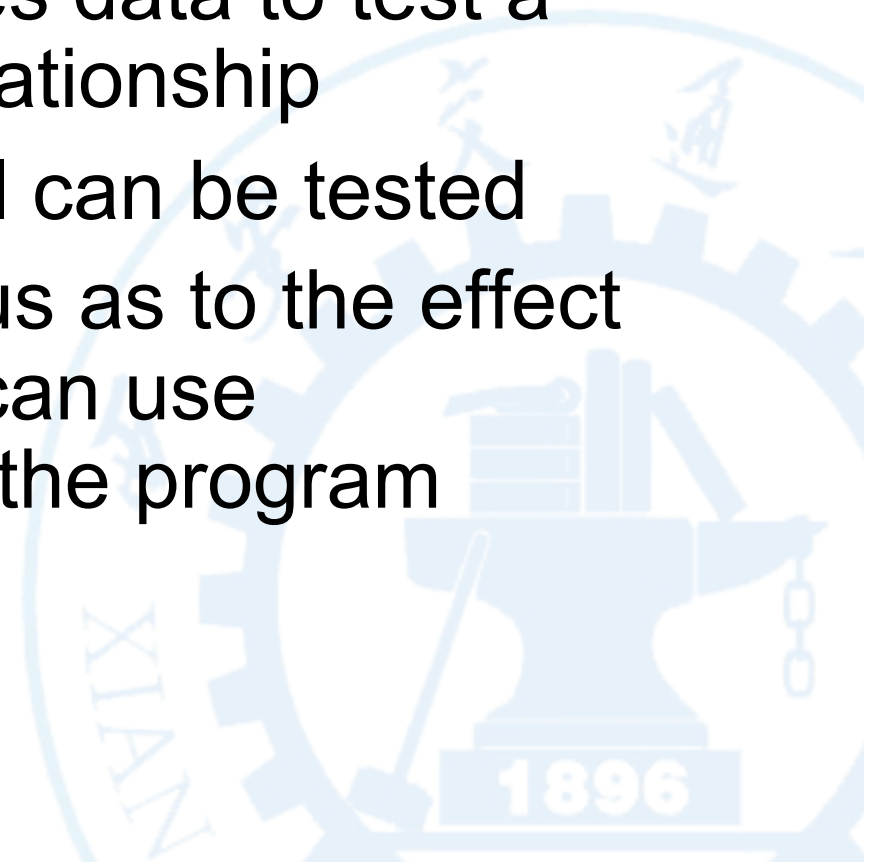
- Rare in economics (and many other areas without labs!) to have experimental data
- Need to use nonexperimental, or observational data to make inferences
- Important to be able to apply economic theory to real world data





Why it is so important?

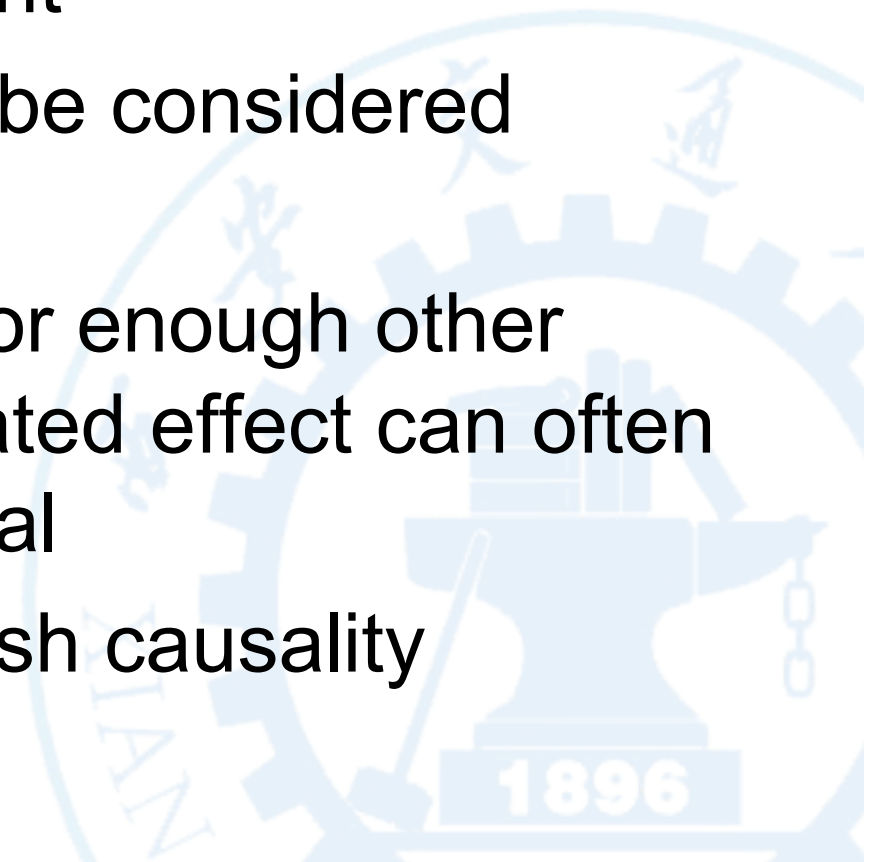
- An empirical analysis uses data to test a theory or to estimate a relationship
- A formal economic model can be tested
- Theory may be ambiguous as to the effect of some policy change – can use econometrics to evaluate the program
- It's a Big Data times.





The Question of Causality

- Simply establishing a relationship between variables is rarely sufficient
- Want to get the effect to be considered causal
- If we've truly controlled for enough other variables, then the estimated effect can often be considered to be causal
- Can be difficult to establish causality





Purpose of Econometrics

- Structural Analysis
- Policy Evaluation
- Economical Prediction
- Empirical Analysis





Methodology of Econometrics

- 1. Statement of theory or hypothesis.
- 2. Specification of the mathematical model of the theory.
- 3. Specification of the statistical, or econometric model.
- 4. Obtaining the data.
- 5. Estimation of the parameters of the econometric model.
- 6. Hypothesis testing.
- 7. Forecasting or prediction.
- 8. Using the model for control or policy purposes.

Example: Kynesian theory of consumption

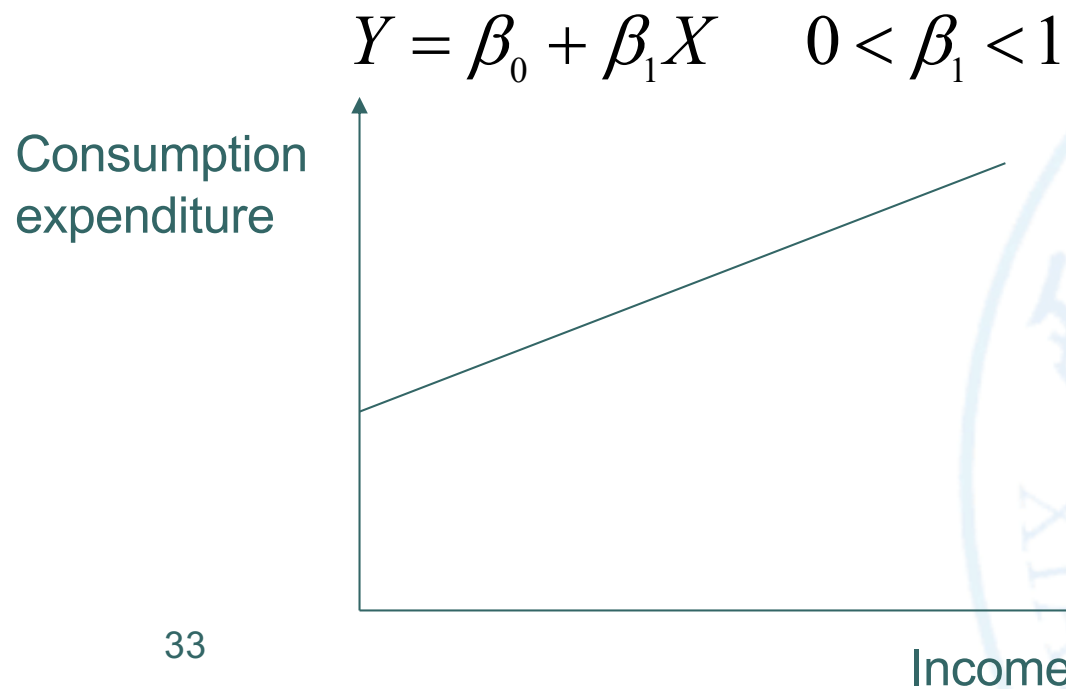
○ 1. Statement of theory or hypothesis.

Keynes stated: The fundamental psychological law is that men/women are disposed, as a rule and on average, to increase their consumption as their income increases, but not as much as the increase in their income.

- In short, Keynes postulated that the marginal propensity to consume (MPC) 边际消费倾向, the rate of change of consumption for a unit change in income, is greater than zero but less than 1

2. Specification of the mathematical model of the theory

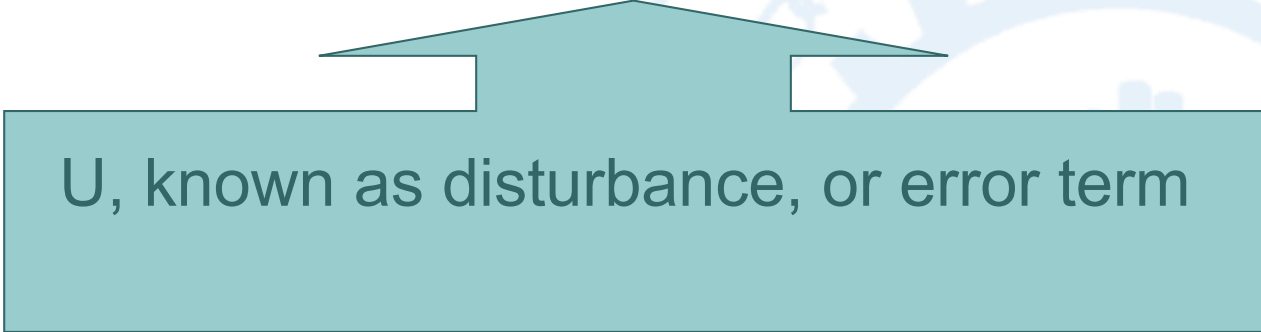
- A mathematical economist might suggest the following form of the Keynesian consumption function:



3. Specification of the statistical, or econometric model.

- To allow for the inexact relationships between economic variables, the econometrician would modify the deterministic consumption function as follows:

$$Y = \beta_0 + \beta_1 X + u$$



U, known as disturbance, or error term

- This is called an econometric model.

4. Obtaining the data.

year	Y	X
1982	3081.5	4620.3
1983	3240.6	4803.7
1984	3407.6	5140.1
1985	3566.5	5323.5
1986	3708.7	5487.7
1987	3822.3	5649.5
1988	3972.7	5865.2
1989	4064.6	6062
1990	4132.2	6136.3
1991	4105.8	6079.4
1992	4219.8	6244.4
1993	4343.6	6389.6
1994	4486	6610.7
1995	4595.3	6742.1
1996	4714.1	6928.4

Source: Data on Y (Personal Consumption Expenditure) and X (Gross Domestic Product), 1982-1996) all in 1992 billions of dollars

5. Estimation of the parameters of the econometric model.

- reg y x

○	Source	SS	df	MS	Number of obs = 15		
○	-----+-----				F(1, 13) = 8144.59		
○	Model	3351406.23	1	3351406.23	Prob > F = 0.0000		
○	Residual	5349.35306	13	411.488697	R-squared = 0.9984		
○	-----+-----				Adj R-squared = 0.9983		
○	Total	3356755.58	14	239768.256	Root MSE = 20.285		
○							
○	-----+-----						
○	y	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
○	-----+-----						
○	x	.706408	.0078275	90.25	0.000	.6894978	.7233182
○	_cons	-184.0779	46.26183	-3.98	0.002	-284.0205	-84.13525
○	-----+-----						

6. Hypothesis testing.

- As noted earlier, Keynes expected the MPC to be positive but less than 1. In our example we found it is about 0.70.
- Then, is 0.70 statistically less than 1? If it is, it may support Keynes's theory.

Such confirmation or refutation of econometric theories on the basis of sample evidence is based on a branch of statistical theory known as statistical inference (hypothesis testing)

7. Forecasting or prediction.

- To illustrate, suppose we want to predict the mean consumption expenditure for 1997. The GDP value for 1997 was 7269.8 billion dollars. Putting this value on the right-hand of the model, we obtain 4951.3 billion dollars.
- But the actual value of the consumption expenditure reported in 1997 was 4913.5 billion dollars. The estimated model thus overpredicted.
- The forecast error is about 37.82 billion dollars.

Using the model for control or policy purposes.

- This is on the opposite way of forecasting.



Example:

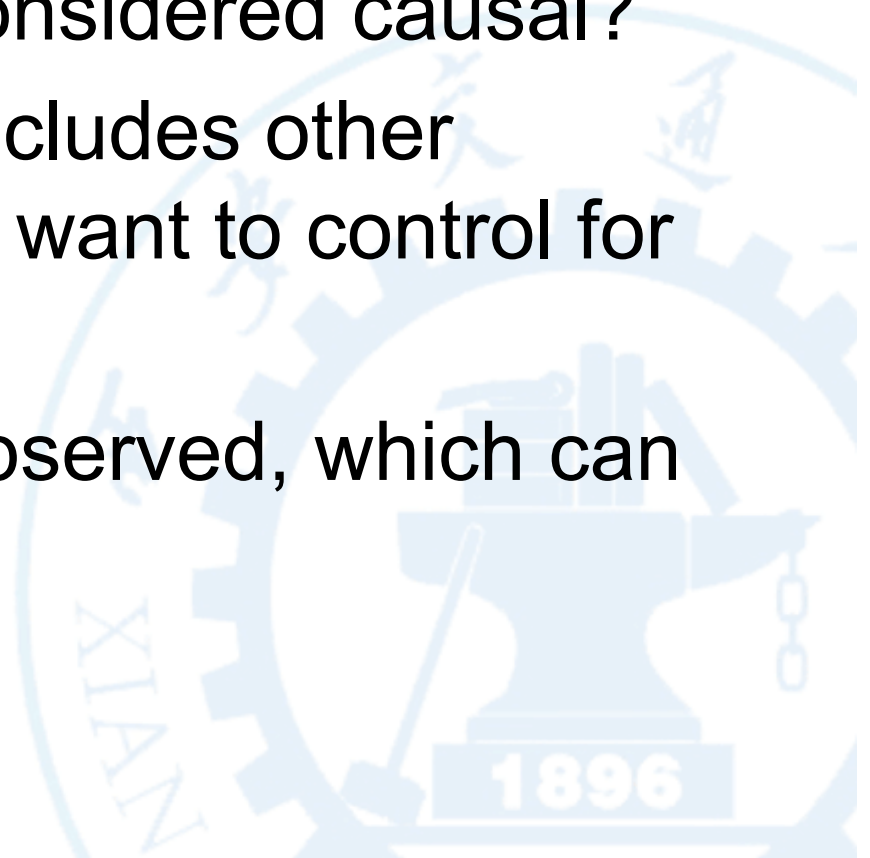
Returns to Education

- A model of human capital investment implies getting more education should lead to higher earnings
- In the simplest case, this implies an equation like

$$Earnings = \beta_0 + \beta_1 education + u$$

Example: (continued)

- The estimate of b_1 , is the return to education, but can it be considered causal?
- While the error term, u , includes other factors affecting earnings, want to control for as much as possible
- Some things are still unobserved, which can be problematic





Types of Data Sets

Cross-Sectional, Time Series, and Panel Data

- Cross-sectional data consist of multiple entities observed at a single time period.
- Time series data consist of a single entity observed at multiple time periods.
- Panel data (also known as longitudinal data) consist of multiple entities, where each entity is observed at two or more time periods.

TABLE 1.1 Selected Observations on Test Scores and Other Variables
for California School Districts in 1998

Observation (district) number	District average test score (fifth grade)	Student-teacher ratio	Expenditure per pupil (\$)	% of students learning English
1	690.8	17.89	\$6,385	0.0%
2	661.2	21.52	5,099	4.6
3	643.6	18.70	5,502	30.0
4	647.7	17.36	7,102	0.0
5	640.8	18.67	5,236	13.9
⋮	⋮	⋮	⋮	⋮
418	645.0	21.89	4,403	24.3
419	672.2	20.20	4,776	3.0
420	655.8	19.04	5,993	5.0

Note: The California test score data set is described in Appendix 4.1.



TABLE 1.1 A Cross-Sectional Data Set on Wages and Other Individual Characteristics

obsno	wage	educ	exper	female	married
1	3.10	11	2	1	0
2	3.24	12	22	1	1
3	3.00	11	2	0	0
4	6.00	8	44	0	1
5	5.30	12	7	0	1
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
525	11.56	16	5	0	1
526	3.50	14	5	1	0





TABLE 1.2 A Data Set on Economic Growth Rates and Country Characteristics

obsno	country	gpcrgdp	govcons60	second60
1	Argentina	0.89	9	32
2	Austria	3.32	16	50
3	Belgium	2.56	13	69
4	Bolivia	1.24	18	12
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
61	Zimbabwe	2.30	17	6



TABLE 1.2 Selected Observations on the Rates of Consumer Price Index (CPI) Inflation and Unemployment in the United States: Quarterly Data, 1959–2000.

Obervation number	Date (Year:quarter)	CPI inflation rate (% per year at an annual rate)	Unemployment rate (%)
1	1959:II	0.7%	5.1%
2	1959:III	2.1	5.3
3	1959:IV	2.4	5.6
4	1960:I	0.4	5.1
5	1960:II	2.4	5.2
⋮	⋮	⋮	⋮
165	2000:II	3.0	4.0
166	2000:III	3.5	4.0
167	2000:IV	2.8	4.0

Note: The U.S. inflation and unemployment data set is described in Appendix 12.1.



TABLE 1.3 Minimum Wage, Unemployment, and Related Data for Puerto Rico

obsno	year	avgmin	avgcov	prunemp	prgnp
1	1950	0.20	20.1	15.4	878.7
2	1951	0.21	20.7	16.0	925.0
3	1952	0.23	22.6	14.8	1015.9
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
37	1986	3.35	58.1	18.9	4281.6
38	1987	3.35	58.2	16.8	4496.7





TABLE 1.4 Pooled Cross Sections: Two Years of Housing Prices

obsno	year	hprice	proptax	sqrft	bdrms	bthrms
1	1993	85,500	42	1600	3	2.0
2	1993	67,300	36	1440	3	2.5
3	1993	134,000	38	2000	4	2.5
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
250	1993	243,600	41	2600	4	3.0
251	1995	65,000	16	1250	2	1.0
252	1995	182,400	20	2200	4	2.0
253	1995	97,500	15	1540	3	2.0
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
520	1995	57,200	16	1100	2	1.5



TABLE 1.3 Selected Observations on Cigarette Sales, Prices, and Taxes,
by State and Year for U.S. States, 1985–1995

Observation number	State	Year	Cigarette sales (packs per capita)	Average price per pack (including taxes)	Total taxes (cigarette excise tax + sales tax)
1	Alabama	1985	116.5	\$1.022	\$0.333
2	Arkansas	1985	128.5	1.015	0.370
3	Arizona	1985	104.5	1.086	0.362
⋮	⋮	⋮	⋮	⋮	⋮
47	West Virginia	1985	112.8	1.089	0.382
48	Wyoming	1985	129.4	0.935	0.240
49	Alabama	1986	117.2	1.080	0.334
⋮	⋮	⋮	⋮	⋮	⋮
96	Wyoming	1986	127.8	1.007	0.240
97	Alabama	1987	115.8	1.135	0.335
⋮	⋮	⋮	⋮	⋮	⋮
528	Wyoming	1995	112.2	1.585	0.360

Note: The cigarette consumption data set is described in Appendix 10.1.



TABLE 1.5 A Two-Year Panel Data Set on City Crime Statistics

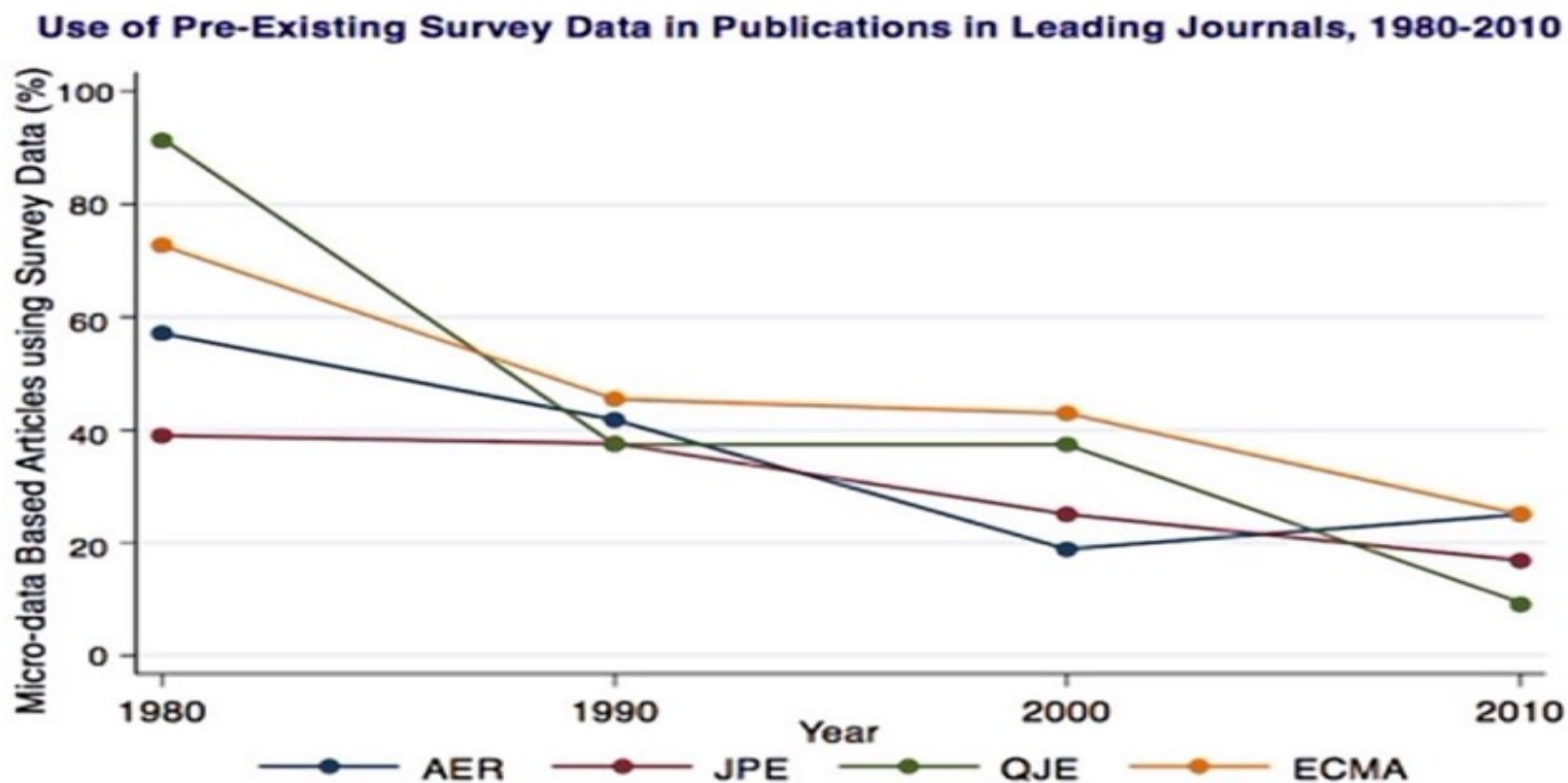
obsno	city	year	murders	population	unem	police
1	1	1986	5	350,000	8.7	440
2	1	1990	8	359,200	7.2	471
3	2	1986	2	64,300	5.4	75
4	2	1990	1	65,100	5.5	75
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
297	149	1986	10	260,700	9.6	286
298	149	1990	6	245,000	9.8	334
299	150	1986	25	543,000	4.3	520
300	150	1990	32	546,200	5.2	493



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Survey Data VS Administrative Data



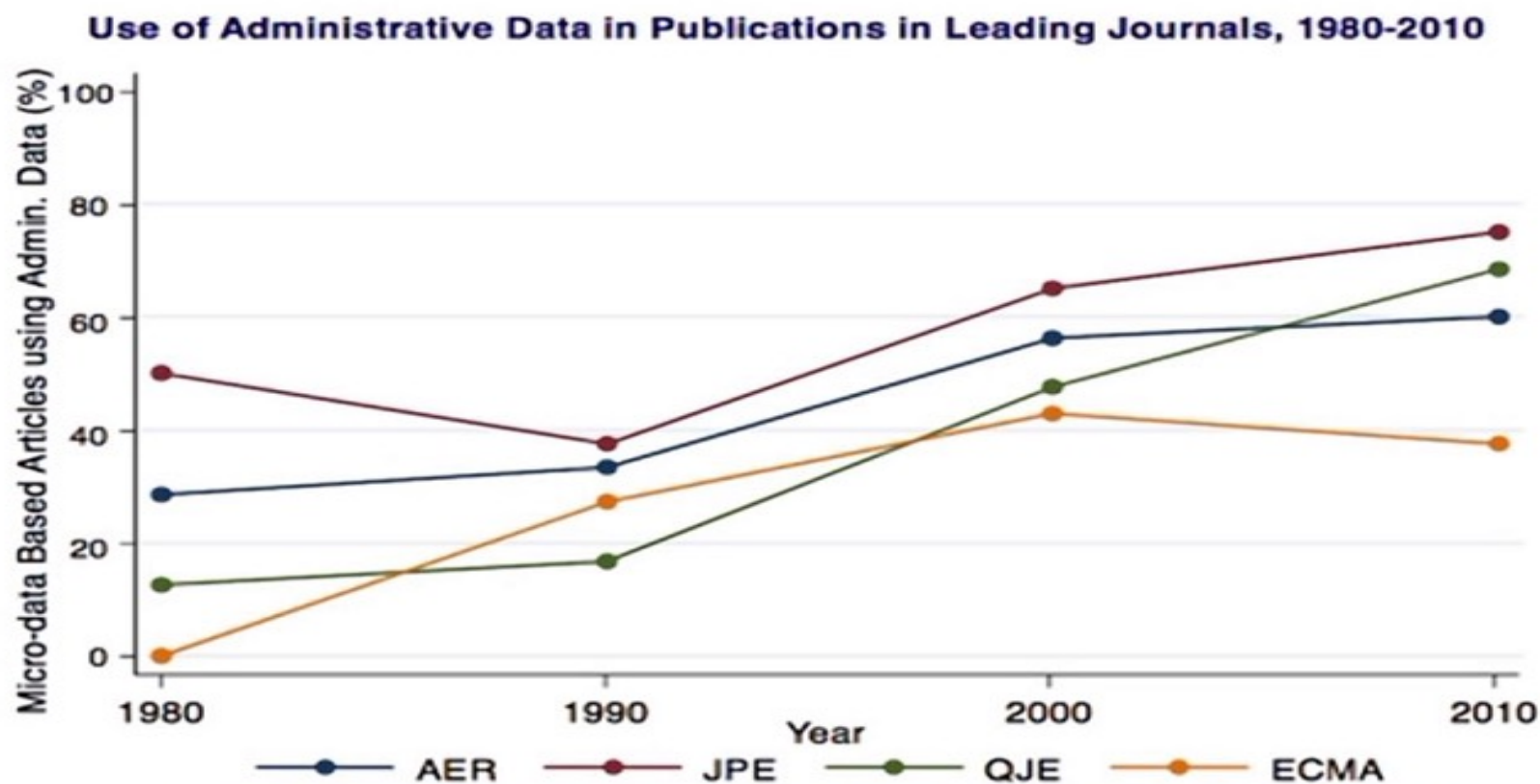
Note: "Pre-existing survey" datasets refer to micro surveys such as the CPS or SIIPP and do not include surveys designed by researchers for their study. Sample excludes studies whose primary data source is from developing countries.



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Survey Data VS Administrative Data



Note: "Administrative" datasets refer to any dataset that was collected without directly surveying individuals (e.g., scanner data, stock prices, school district records, social security records). Sample excludes studies whose primary data source is from developing countries.



Homework

- Review the basic statistics
- Read through Chap. 1

